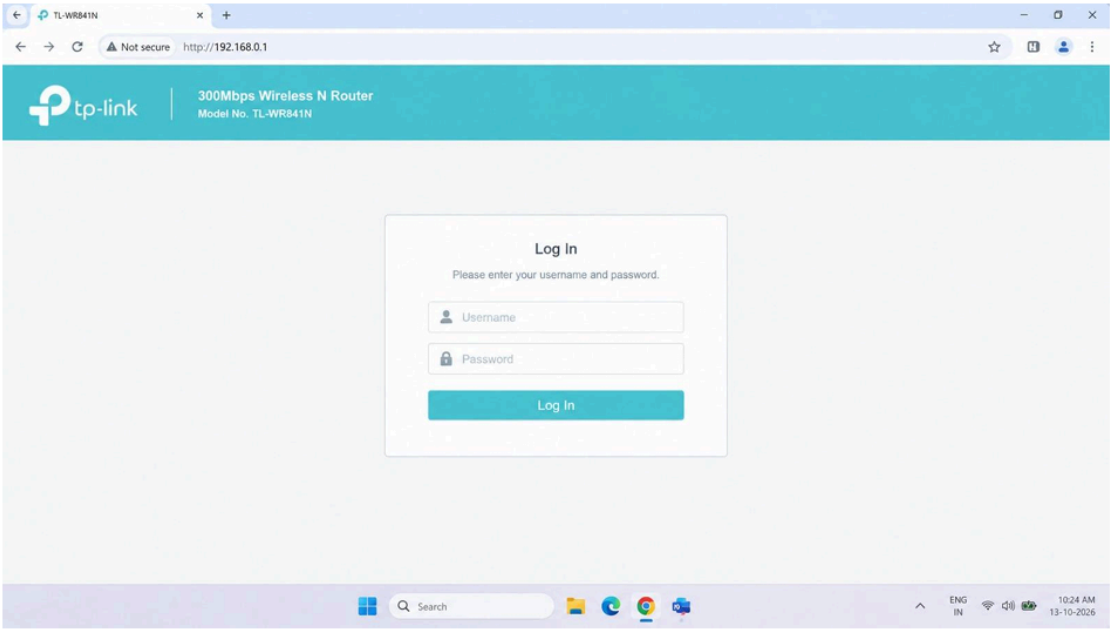
Assignment-4.3

“IP address and configuration”

TASK-1.Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

Steps to Open the Router Login Page: -
• Connect your laptop to your home Wi-Fi.
• Press Windows + R, type cmd, and press Enter.
• Type this command and press Enter: ipconfig
• Find Default Gateway under your Wi-Fi adapter. For example, it may be 192.168.1.1.
• Open Chrome and type that IP address in the address bar and type your IP in your chrome.
• The router admin login page should open. Take a screenshot using Windows + Shift + S.



TASK-2 Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network

IPv4 Address. . . . . : 10.230.50.149
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :10.230.50.116

TASK-3 Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.Remember to revert to automatic/DHCP after testing to avoid network issues

Objective

To change the laptop's IP address from automatic (DHCP) to static and test internet connectivity.

Current Network Details

- IP Address: 10.230.50.149
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.230.50.116

Procedure

1. I checked my laptop's network details using the `ipconfig` command in Command Prompt.
2. I opened Windows Settings and went to Network & internet → Wi-Fi.
3. I opened my connected Samsung phone's hotspot properties and selected Edit under IP assignment.
4. I selected Manual and enabled IPv4.
5. I entered a suitable static IP address, subnet mask, gateway, and DNS settings.
6. I tested the internet connection by opening a website.
7. After testing, I restored the IP assignment to Automatic (DHCP).

Result

The laptop successfully connected to the internet after changing the IP address to static. After testing, I restored the IP settings to Automatic (DHCP).

TASK-4 In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

DHCP Range: - Current DHCP Range: Not available because I have not accessed the actual router's admin panel. You can use this sample range for a theory assignment.
• Example Start IP: 192.168.1.100
• Example End IP: 192.168.1.200

What happens if two devices get the same IP address? If two devices are assigned the same IP address, an IP conflict occurs. Both devices may face network connection problems, and internet access may become unstable. The router should assign unique IP addresses to avoid this problem.